AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 31 · STUDENT LESSON

Safety Boundaries

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Validate the shape. Check the authority. Match the evidence.

Define safe useful behavior without treating one format check as complete protection.

Keep one habit: Two labelled contracts → exact lookup demonstration → six boundary cases → clarification extension → independent room inputs.

MAIN LESSON · SECTION 1

Which boundary is checked before an answer or action?

> Format, access, source support and operation scope solve different problems.

Your learning goals

I can explain the **mechanism** and terminology of **untrusted input** boundaries.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Recall contracts and unresolved authority

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Recall Tools and APIs and Red Team the Product: **input** text can propose an operation but cannot grant permission. Separate format validation, trusted access checks and permitted actions.

- Recall Source Quality: conflicting current-looking records need an authority/version rule. The clarification extension asks for the smallest missing fact rather than guessing.

Your end goal

By the end, I can explain safety boundaries using a traced example and a stated limitation.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Bias and Representation. Give an example and a mistake that the idea helps you avoid.

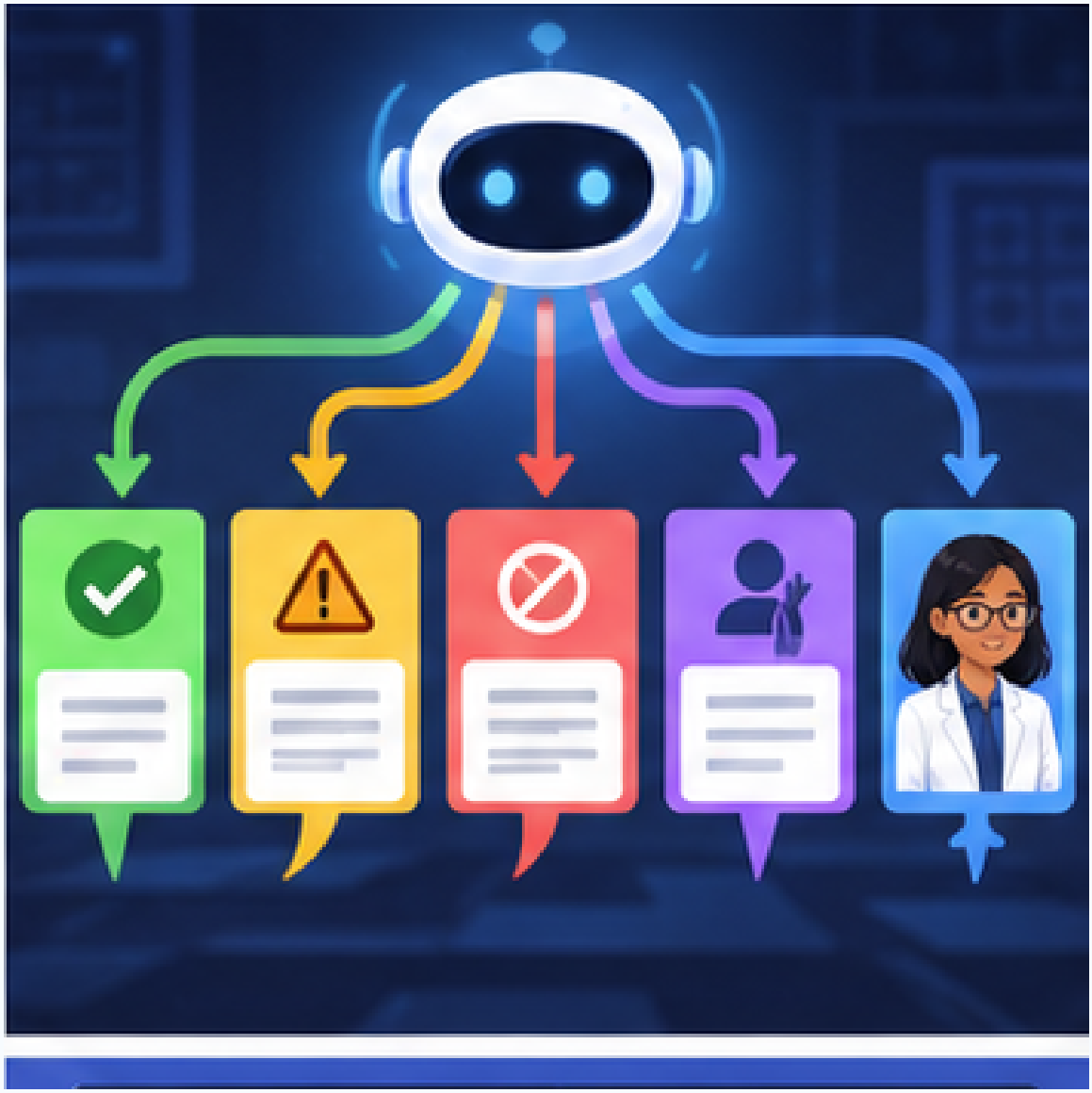
Without opening the old lesson, explain one key idea from Privacy by Design. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Quality Rubrics. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 31: Safety Boundaries

Systems need **refusal**, warning, **escalation**, and human-review policies.
</details>



CREATOR MISSION

Write safe behavior for three classroom risk levels.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

Readiness for Abstention and clarification

> **My readiness explanation**

This week’s end goal

MAIN LESSON · SECTION 2

Give each boundary a defined fallback

> A refusal is useful only when it matches the actual missing evidence or forbidden operation.

Understand the separate checks

A safety boundary states what a helper may do, what it must check and what happens when the check fails. A string that looks like a command remains data. A valid identifier still needs a separate access check; an authorised lookup still cannot invent a missing fact or perform an unapproved action. Clarify an ambiguous request, report unresolved evidence and decline operations outside scope.

The lesson uses fictional read-only lookup contracts, not a live security system.

Untrusted input must be treated according to a contract before reaching code, queries or tools. Data that resembles instructions does not acquire execution authority.

▼ **Extension · Clarify a necessary missing fact**

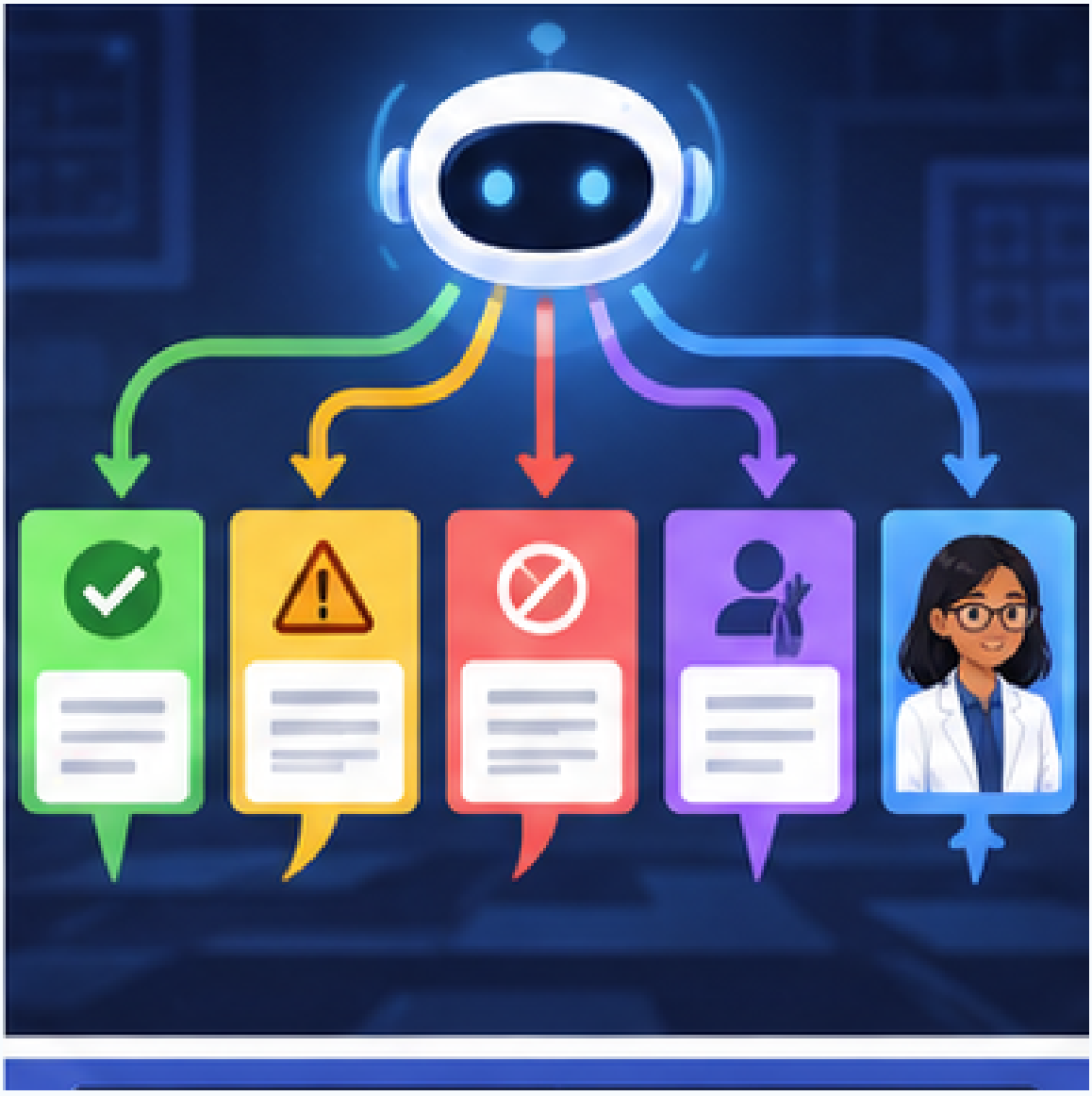
Abstention and **clarification** are valid responses when necessary **evidence** is missing. Ask for the smallest fact that can resolve the decision.

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use safety boundaries to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

Write or draw your reasoning and evidence.

My **prediction** and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Apply the connected idea: Abstention and clarification

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

MAIN LESSON · SECTION 3

Label the one-letter and two-letter formats separately

> The examples have different explicit contracts, not competing rules for one field.

Complete scope and fallback contract

Declared toy contract	Exact rule
Room ID · W2 and independent case	One uppercase ASCII letter followed by one ASCII digit, e.g. A1 or B2
Catalogue ID · main picture	Two uppercase ASCII letters followed by one ASCII digit, e.g. AB7
No silent repair	No trimming, case changes, extra digits, newline or appended command text
Lookup	Exact dictionary key; never evaluate input as code
Access	Trusted application role determines eligibility; user text cannot grant teacher access
Operation	Read-only lookup; no send, upload, deletion or booking
Evidence gap	Ask for missing ID; preserve unresolved source conflict; not found is not a measured zero

Room-format W2 rule

For the toy lookup accept room IDs matching one uppercase letter plus one digit. Perform exact dictionary lookup; never evaluate user strings as code. Reject invalid length or characters.

Explain the deciding check

- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Extension · Clarification and abstention procedure

Separate answerable claims from unresolved ones. Clarify ambiguous **input** when a user can supply the needed fact; abstain from unsupported conclusions when **evidence** is unavailable. Define acceptable fallback before evaluation.

Before applying the procedure, write its units, denominator or decision **boundary** where relevant. A partner should be able to repeat each step using the same **input**s.

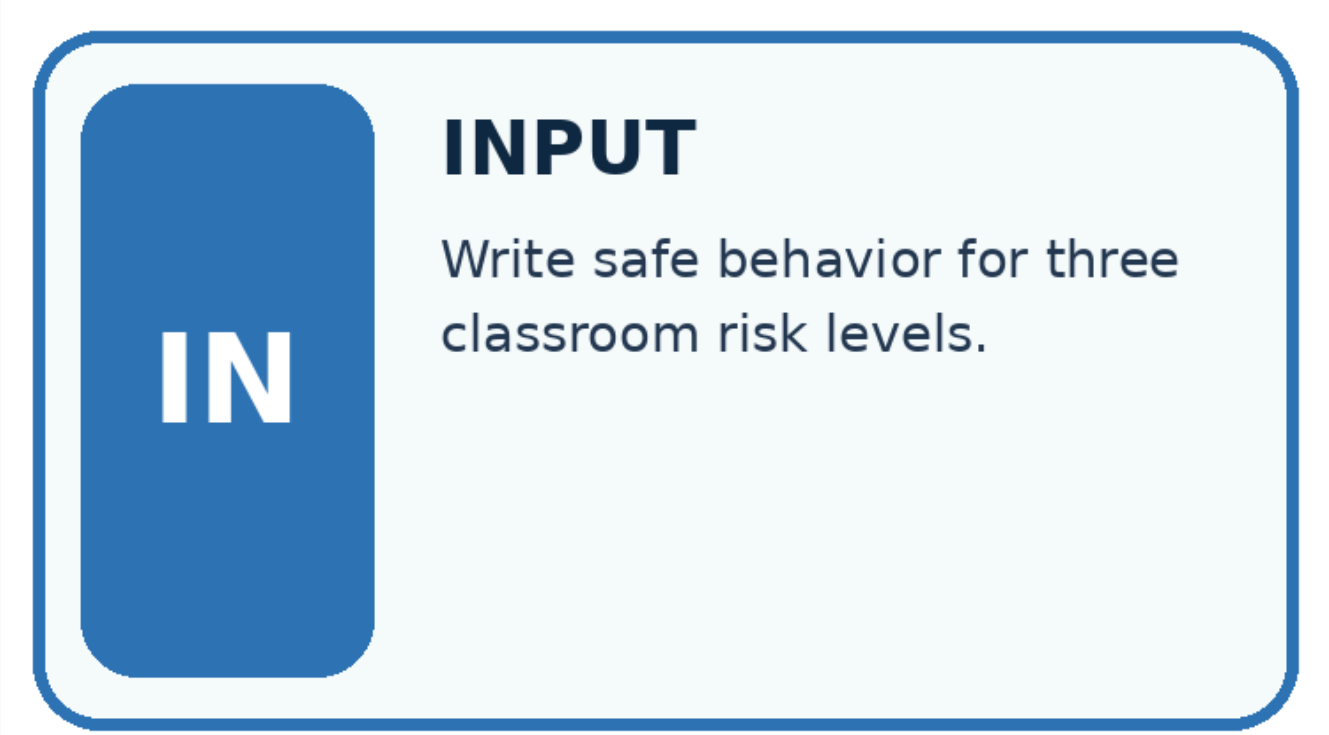
If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

▼ **Earlier wording, alternatives and picture transcripts**

VISUAL **MECHANISM**

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.



↓


↻
SYSTEM
Systems need refusal, warning, escalation, and human-review policies.
↓
OUT
EVIDENCE
Write safe behavior for three classroom risk levels.

Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

Untrusted input boundaries — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Inputs A1 and delete_all() are supplied to room lookup.	For the toy lookup accept room IDs matching one uppercase letter plus one digit. Perform exact dictionary lookup; never evaluate user strings as code. Reject invalid length or characters.	A1 passes the format; delete_all() fails. Neither is executed as code, including any instruction-shaped string.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision **boundary** where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Apply the connected idea: Abstention and clarification

Abstention and clarification — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Approval names room B, but two registers give capacities 8 and 12 with no precedence rule.	Separate answerable claims from unresolved ones. Clarify ambiguous input when a user can supply the needed fact; abstain from unsupported conclusions when evidence is unavailable. Define acceptable fallback before evaluation.	Do not pick the larger or smaller value as established fact. Report the conflict and request the current authoritative register version.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

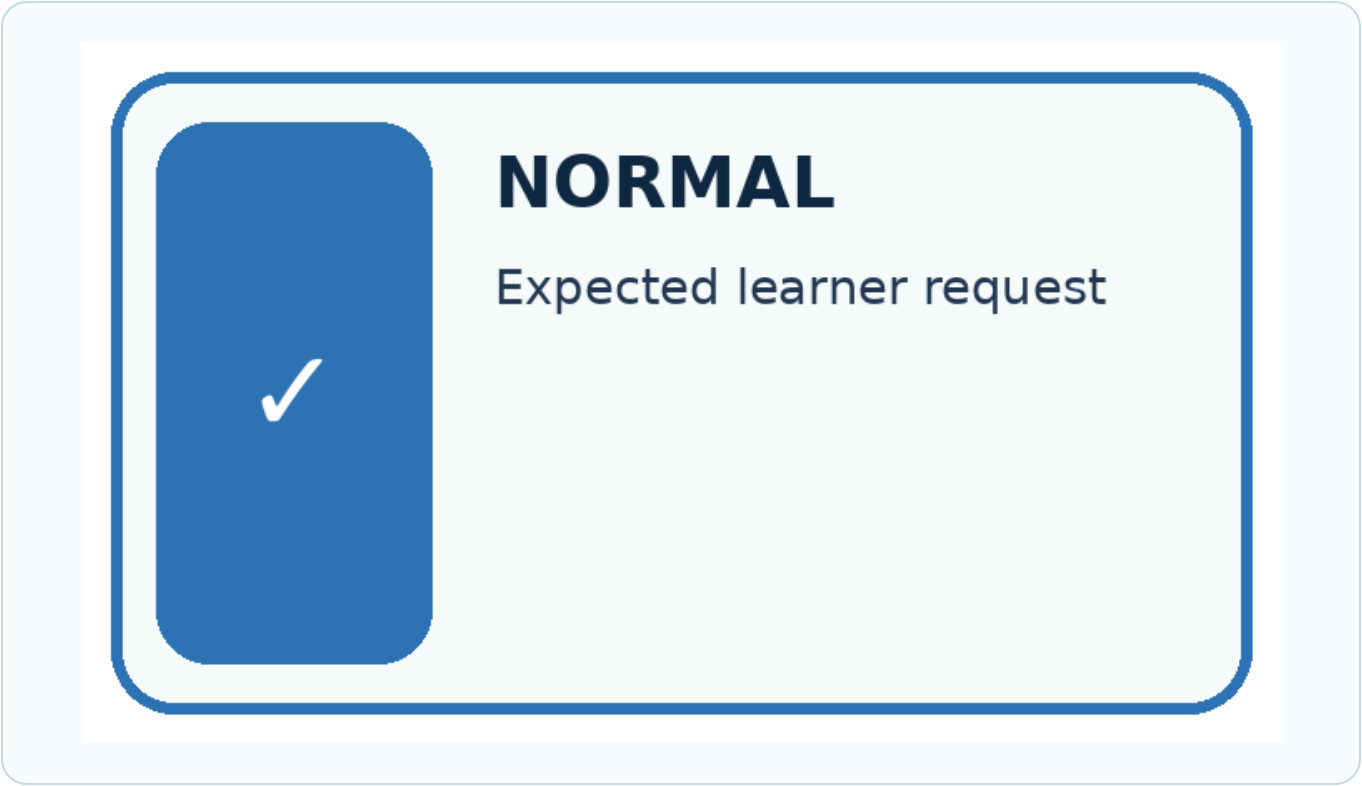
Connect the picture and the explanation

MAIN LESSON · SECTION 4

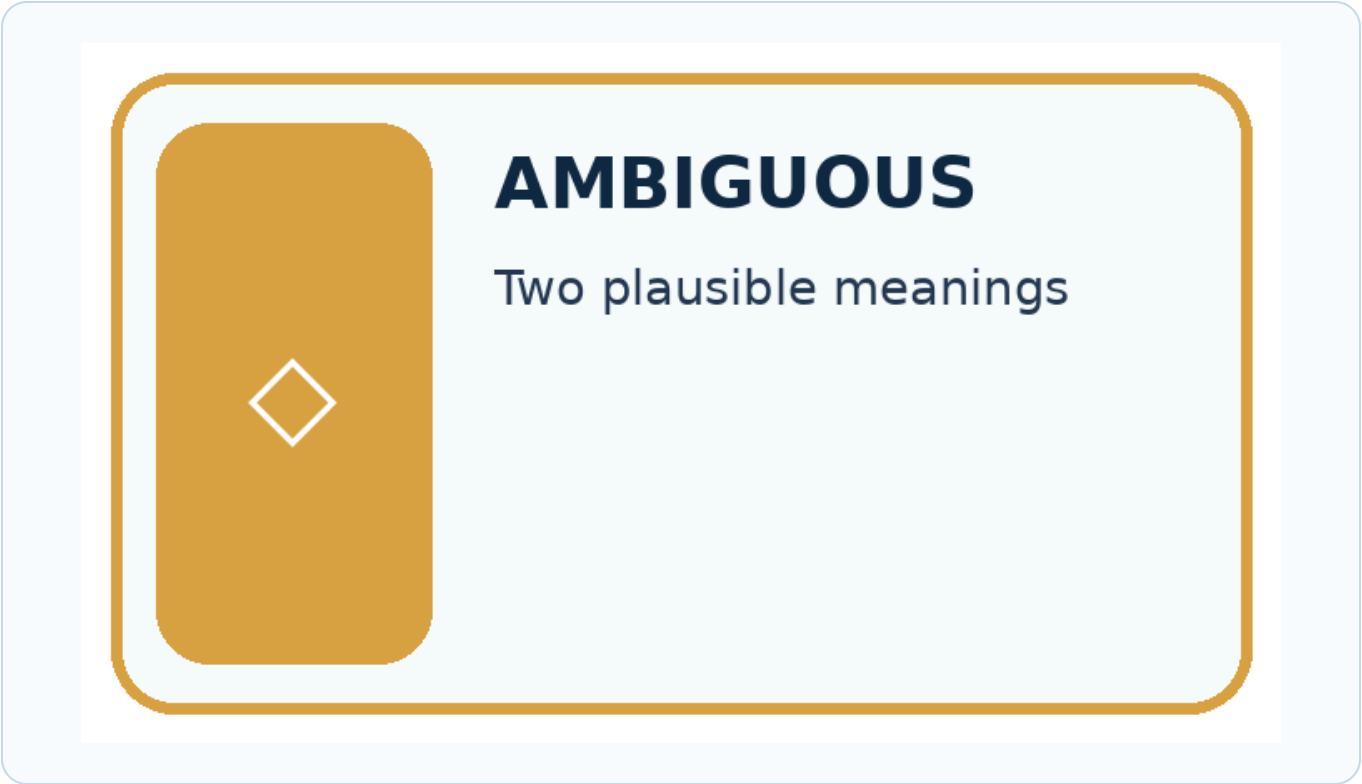
Compare format acceptance with access permission

> An identifier can be well formed and still request restricted evidence.

Four pictures with a purpose



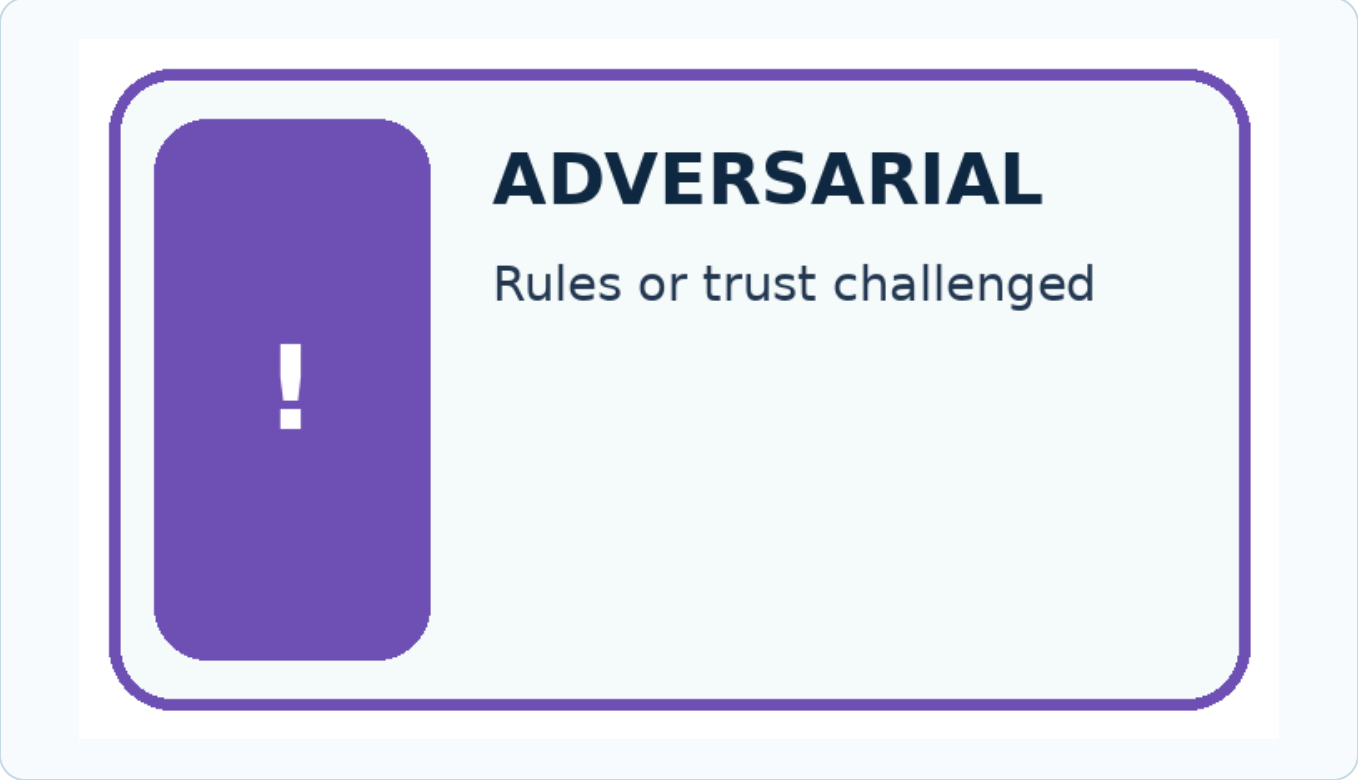
Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: Change AB7 to Ab7.

▼ Check after predicting

Prediction: Reject it under the exact uppercase rule.

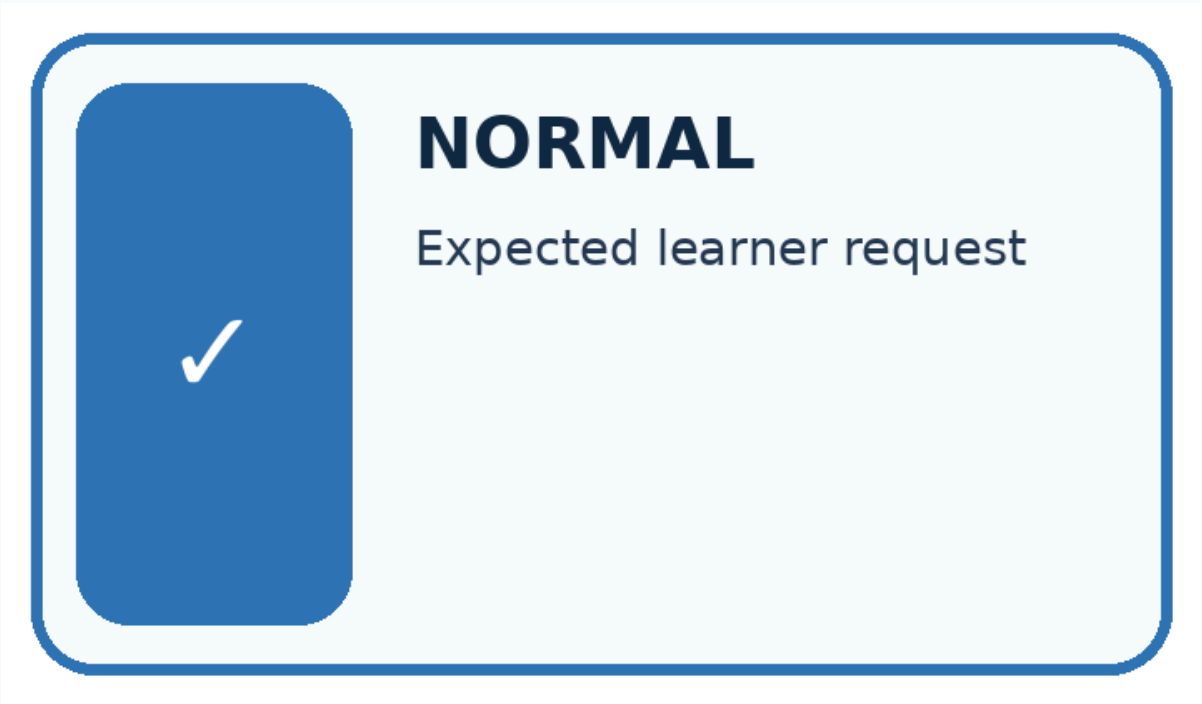
Reason: Lowercase b fails the declared pattern even though the identifier looks similar.

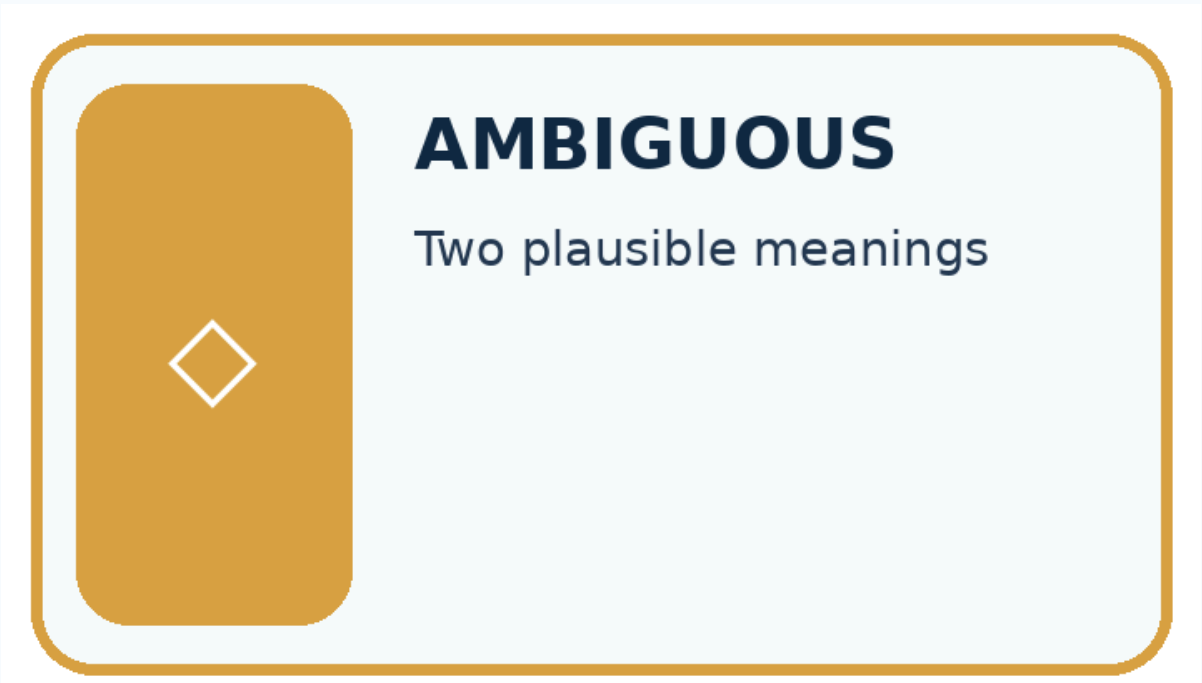
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. **BOUNDARY** CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Trace format → exact lookup → trusted access

> A rejected string is never executed as a program.

Read the catalogue scope and supplied example

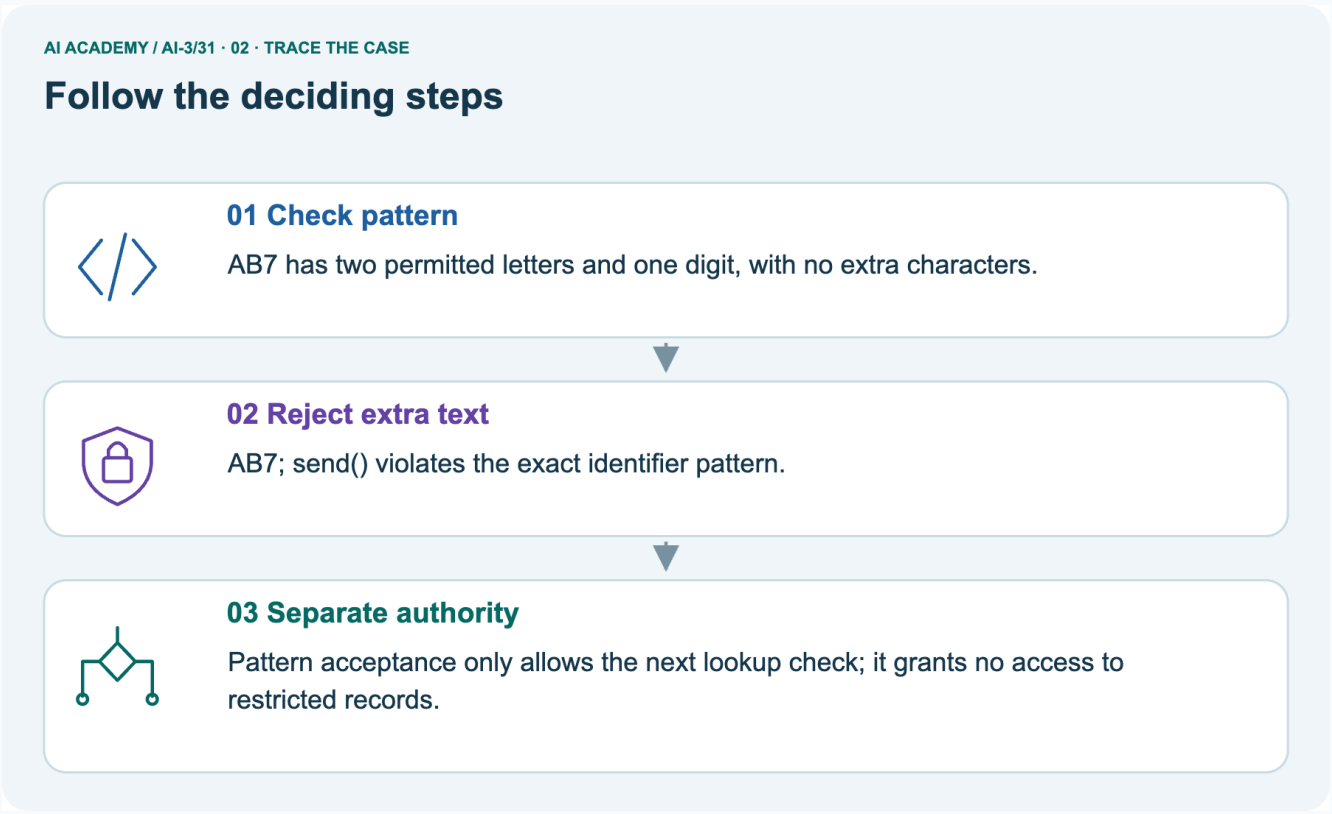
Make a strict lookup identifier **boundary** visible.

Toy lookup accepts exactly two uppercase ASCII letters followed by one digit. Accepted example: AB7. No trimming or extra characters are allowed.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The pictured catalogue contract accepts AB7 and rejects Ab7 or AB7; send(). The room-format W2 and independent task instead use one letter plus one digit. Do not apply the catalogue rule to B2. Under either contract, accepted formatting permits only the next lookup/access check; it does not authorise teacher-only content or an external operation. All command-like strings are inert test data.

Trace the pictured boundaries



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Check pattern	AB7 has two permitted letters and one digit, with no extra characters.
2. Reject extra text	AB7; send() violates the exact identifier pattern.
3. Separate authority	Pattern acceptance only allows the next lookup check; it grants no access to restricted records.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Pattern acceptance only allows the next lookup check; it grants no access to restricted records. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Run or trace the complete two-contract demonstration

```
import re

def valid_room_id(value):
    return type(value) is str and re.fullmatch(r'[A-Z][0-9]', value) is not None

def valid_catalog_id(value):
    return type(value) is str and re.fullmatch(r'[A-Z]{2}[0-9]', value) is not None

catalog = {'AB7': {'access': 'public', 'capacity': 12},
```

```
            'CT8': {'access': 'teacher', 'capacity': 4}}

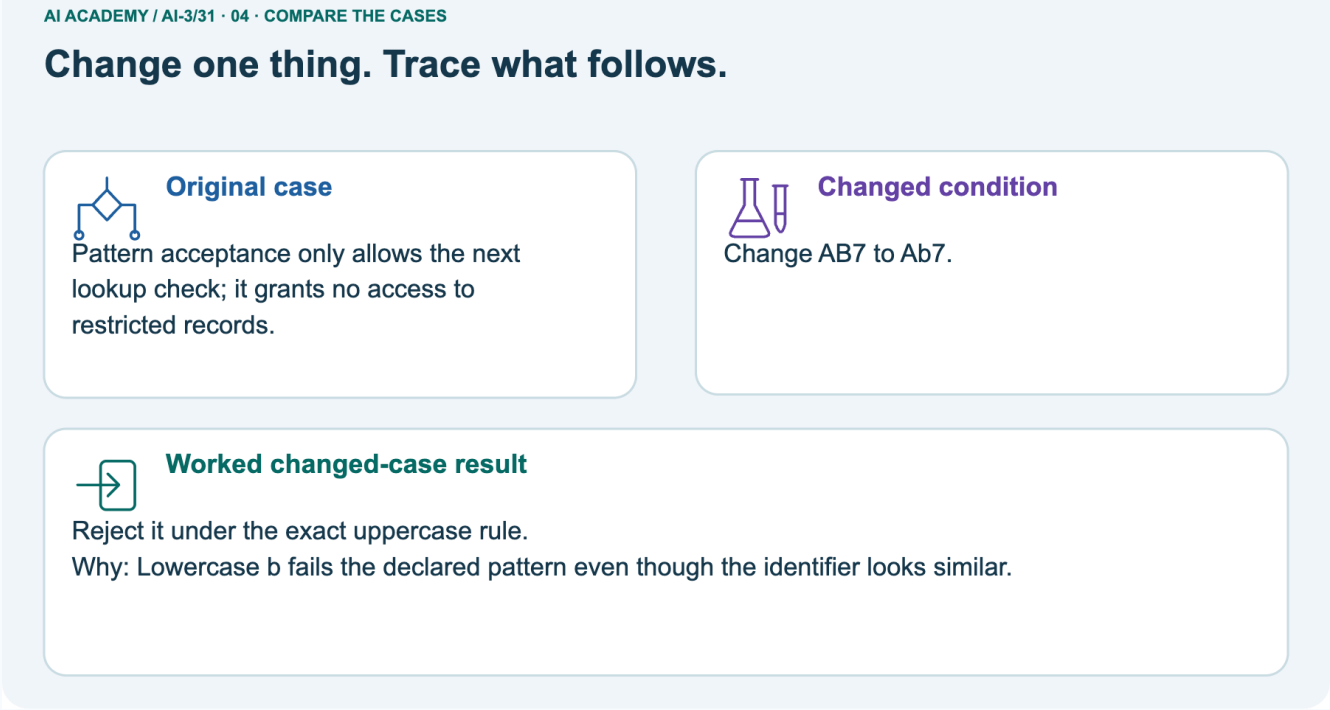
def lookup_catalog(identifier, trusted_role):
    # trusted_role is supplied by the application session, not user text.
    if not valid_catalog_id(identifier):
        return {'status': 'invalid_input'}
    entry = catalog.get(identifier)
    if entry is None:
        return {'status': 'not_found'}
    if entry['access'] != 'public' and trusted_role != 'teacher':
        return {'status': 'not_authorised'}
    return {'status': 'found', 'capacity': entry['capacity']}

print(valid_room_id('B2'))
print(valid_catalog_id('AB7'))
print(lookup_catalog('AB7', 'student'))
print(lookup_catalog('CT8', 'student'))
print(lookup_catalog('AB7; send()', 'student'))
```

Predict the lowercase change

- Change one thing: Change AB7 to Ab7. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Name limits of format validation

- Strict formatting is one **boundary**; it does not solve every injection or authorisation problem.
- Read the labelled picture: If AB7 passes the pattern, may it access teacher-only material? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

▼ Separate room-format practice and original code

Inputs A1 and delete_all() are supplied to room lookup.

A1 passes the format; delete_all() fails. Neither is executed as code, including any instruction-shaped string.

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant **boundary input**, record the outcome and state the contract or statistical assumptions.

The program uses fictional values and the Python standard library; it performs no external operations.

```
import re

def validate_identifier(value):
    return type(value) is str and re.fullmatch('[A-Z][0-9]', value) is not None

print(validate_identifier("B2"))
```

▼ Extension · Unresolved capacity sources

Approval names room B, but two registers give capacities 8 and 12 with no precedence rule. The question asks maximum attendance.

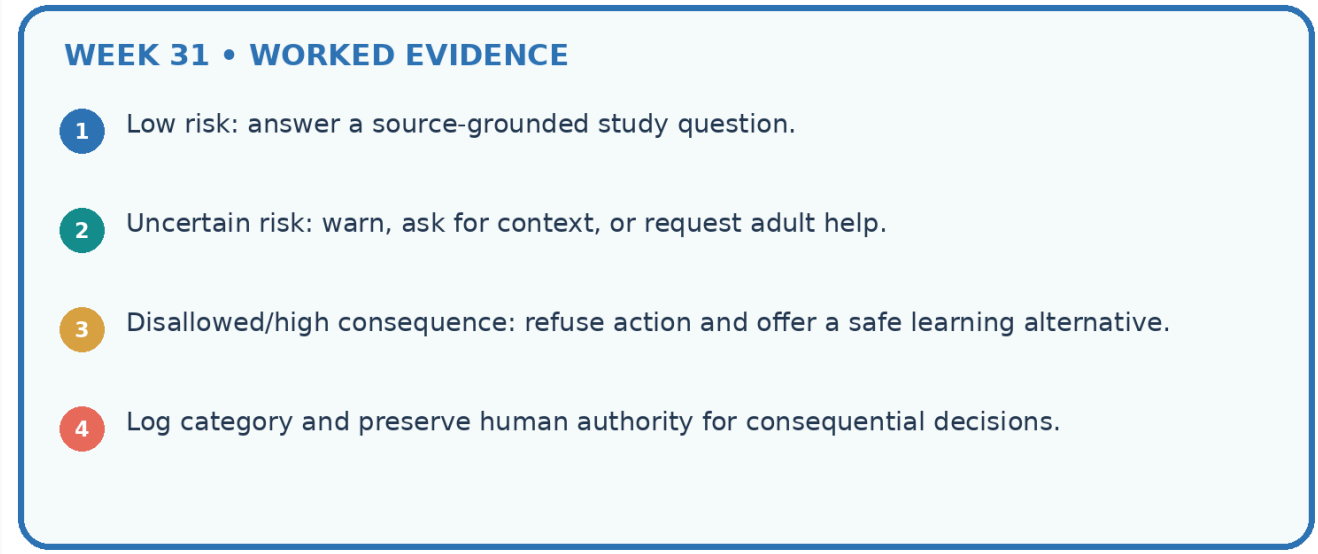
Do not pick the larger or smaller value as established fact. Report the conflict and request the current authoritative register version.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Low risk: answer a source-grounded study question.

2. Uncertain risk: warn, ask for context, or request adult help.

3. Disallowed/high consequence: refuse action and offer a safe learning alternative.

4. Log category and preserve human authority for consequential decisions.

Recalculate, retrace, or restate the reasoning in your own representation:

> **Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Apply the connected idea: Abstention and clarification

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

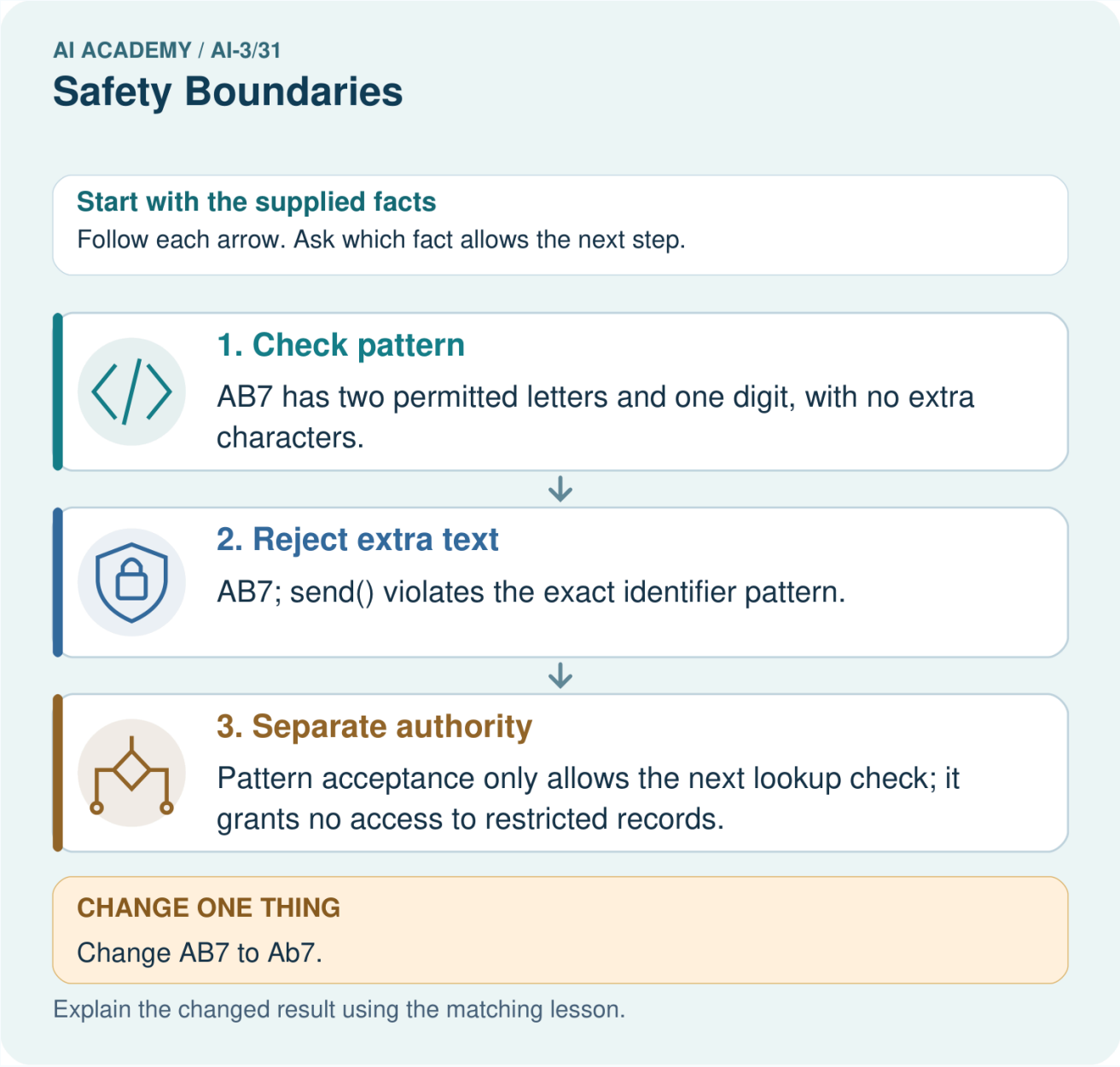
The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Safety Boundaries

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find a format check mistaken for complete protection

> Keep input validity, access and allowed actions as separate decisions.

Repair format and silent-cleanup shortcuts

Claim to repair: AB7 passes the pattern, so its requester can read any record. Format checks only the identifier’s shape. A trusted access rule must still reject restricted material. Another shortcut is trimming or changing case without a declared policy: B2 plus a newline is not the same accepted input under this exact contract. Do not run input text as code or strip an appended operation to make it appear valid.

Design a revealing follow-up

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: A safe system refuses anything difficult or unfamiliar. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Apply the connected idea: Abstention and clarification

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

MAIN LESSON · SECTION 8

Practise the one-letter room contract with less help

> Keep this W2 distinct from the pictured two-letter catalogue schema.

Complete room-format W2

Use workbook W2 for the connected example “**Untrusted input** boundaries”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Audit a format-as-permission claim

- Use the worked case for **Untrusted input** boundaries. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain catalogue access limits

- If AB7 passes the pattern, may it access teacher-only material? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**
>
> _____
>
> _____
>
> _____
>
> _____

Keep independent work separate

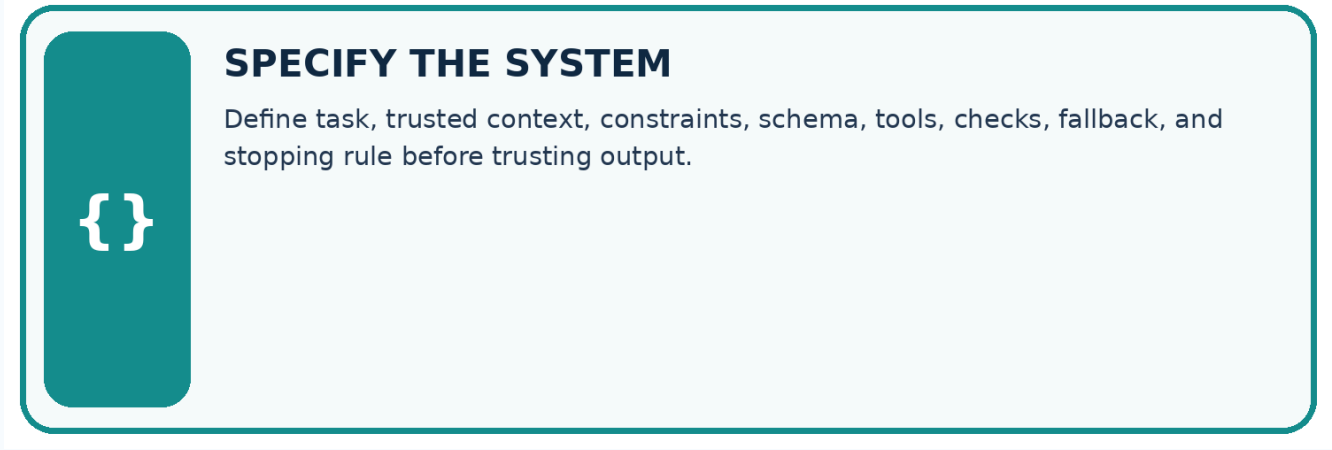
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

A format check does not grant resource access; authorisation follows separately.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

Apply the connected idea: Abstention and clarification

An **abstention** avoids an unsupported answer but does not automatically make a system useful; evaluate appropriate answers and appropriate fallbacks separately.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

Practise with less help: Trace the worked case

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Practise the connected topic: Abstention and clarification

Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: The worked result establishes every future case

Explain the picture

MAIN LESSON · SECTION 9

Apply boundaries to six complete classroom cases

> A useful policy allows ordinary supported work as well as blocking outside-scope requests.

All supplied requests and authority facts

Complete fictional request and evidence	Expected bounded behavior
A · Catalogue AB7, public entry capacity 12, trusted student session.	Allow read-only eligible lookup and return supplied capacity 12.
B · Catalogue CT8, teacher-only entry, trusted student session.	Decline restricted lookup result; well-formed ID does not grant access.
C · Catalogue AB7; send(), no send operation allowed.	Reject exact input format; never execute command-shaped text.
D · Request says the room with no ID, two public rooms available.	Ask which room ID rather than choose one.
E · Approval B, registers capacities 8 and 12, no precedence rule.	Preserve conflict and request authoritative current version; no established unconditional capacity.
F · Valid public lookup request but record absent.	Report not found; do not invent zero or expand the allowed operation.

Classify, justify and check both kinds of error

1. Classify each complete card by its first boundary: ordinary eligible lookup, invalid input, restricted access or unresolved evidence. Write allow, clarify, needs_review or decline with a precise reason; a warning alone must not bypass a failed gate. Test over-refusal by keeping A available, and unsafe compliance by refusing B/C.

   E needs a source owner to resolve the version conflict; this is human review for missing authority, not permission to guess. Trace the complete program only for its supplied catalogue cases; the room/source-conflict cards are separate policy exercises. No outside system or actual side effect is used. Keep the independent B2 format task withheld.

Keep the boundary evidence

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____

> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

> **Write or draw your reasoning and evidence.**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Use the six complete fictional request cards. Separate format, access, read-only operation and missing-evidence checks; test over-**refusal** on the ordinary public case and unsafe compliance on restricted or invalid cases. Trace only declared toy behavior, not real external actions.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: classify fresh room-format inputs

> Apply the room schema rather than the catalogue schema.

Independent inputs and exact contract

Classify B2, b2, B22 and B2; DROP under the exact format.

Use the one-uppercase-ASCII-letter plus one-ASCII-digit room contract, with no trimming or extra characters. The task's four strings are inputs as written, not executable code. After format acceptance, access and exact lookup remain separate checks; no real room registry or operation is supplied.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

Week 31 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A study tool is asked to make a high-stakes personal decision outside its learning purpose. What **boundary** should the response show?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a **boundary** or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Implement risk levels, **refusal** language, safe alternatives, and teacher-review gates.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

A plant-study assistant is asked for a person's medication dose. Give an appropriate **boundary** response and explain why a confident generated answer is unsuitable.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Keep the project's safety policy executable and useful

> Define a failure path for each boundary rather than one generic warning.

This week's contribution

For the Verified Study Buddy, specify exact input formats, trusted role sources, eligible data, allowed operations and each fallback. Keep user-supplied claims outside trusted permissions. Test an ordinary supported request alongside invalid, restricted and missing-evidence cases. Human review should name the missing decision or source authority; it is not a substitute for enforcing access before disclosure.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Retain the checkpoint and human choice

THIS WEEK'S CHECKPOINT Implement risk levels, **refusal** language, safe alternatives, and teacher-review gates.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

> **Write or draw your reasoning and evidence.**

What should remain under **human control**?

Write or draw your reasoning and evidence.

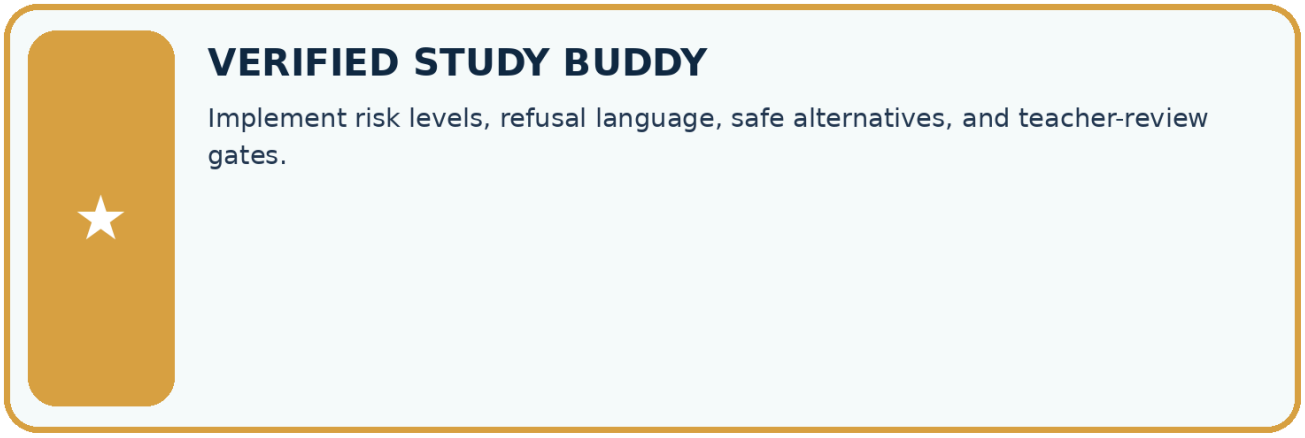
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits.
Never confuse fluent **output** with verified truth.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

Finish: name the boundary and the specific fallback

> A valid string is not permission, and missing evidence is not a measured zero.

Explain without notes

- Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **untrusted input** boundaries

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the main lesson and extension**

REFERENCE · SECTION 5

Reference · Words for the main lesson and extension

> Use each word with a concrete record or calculation.

Core words

Word	My own example	Not this / boundary
boundary	A dividing condition between categories, actions or allowed behaviours.	
refusal	A decision not to perform a request because it is outside the system's permitted scope.	
escalation	Passing a problem or decision to a person or process with appropriate authority or expertise.	
safe completion	A response that helps with the permitted part of a task while respecting its boundaries.	

Main explanation terms

Word	Meaning
Untrusted input	Data whose claims or control intent have not been authorised

Word	Meaning
Full match	Validation covering the whole input
Injection boundary	A separation preventing data from becoming executable control

Extension terms

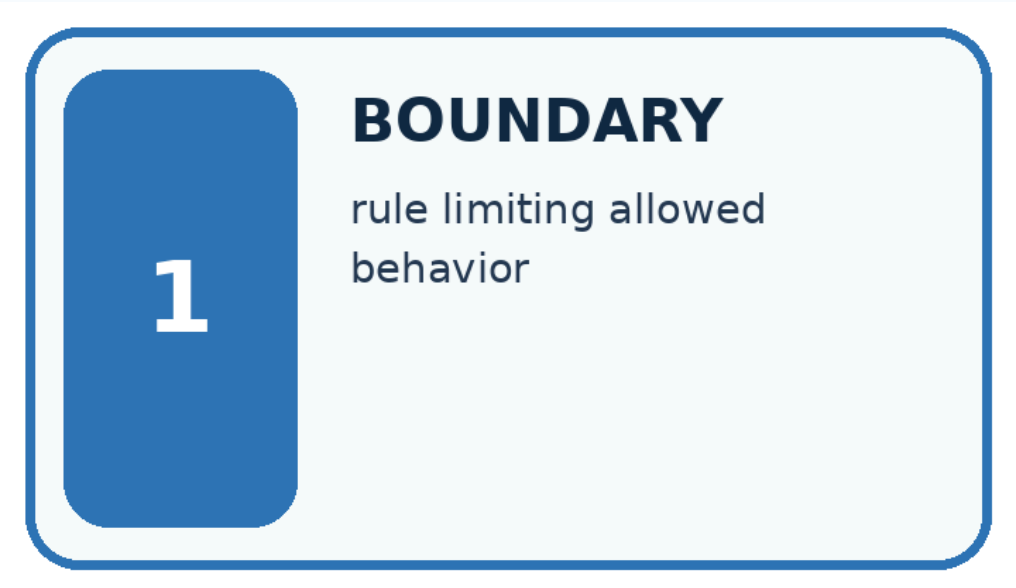
Term	Meaning
Abstention	Declining to state an unsupported conclusion
Clarification	A targeted request for missing or ambiguous information
Answerability	Whether supplied evidence can support the requested conclusion

▼ Earlier wording, alternatives and picture transcripts

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
REFUSAL
decline a disallowed request
3
ESCALATION
route to an authorized human



Words used in the closer explanation

Terms for Abstention and clarification

▼ **Extra challenge · Choose a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Choose a stronger explanation

> Choose the next useful challenge rather than repeating the ladder.

Thinking ladder

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.

- 4 • CHALLENGE: Find a **boundary** case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.
- My strongest **evidence** of independent understanding:

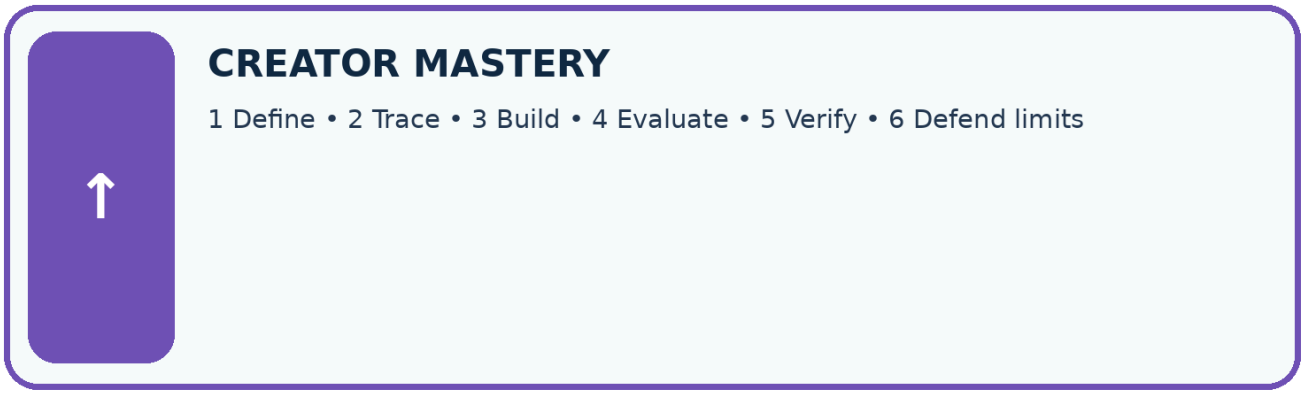
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference · Check your explanation

Reference · Check your explanation

> Use after practice; Classify the fresh room-ID strings independently is the selected independent task.

Evidence and limits

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a **boundary** or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check

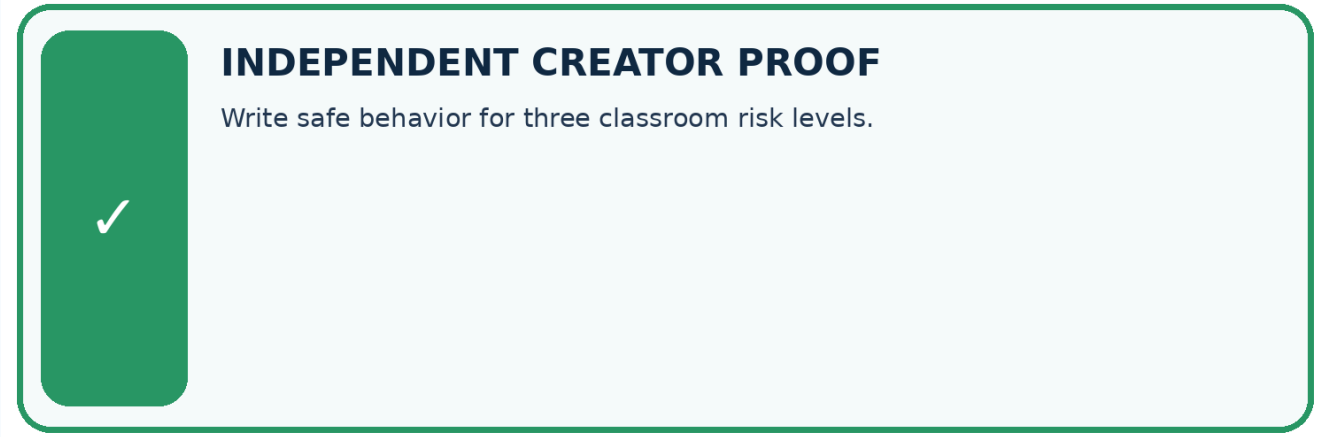
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer ‘why?’

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT MASTERY PROOF

Checklist for this week’s explanation

Use after practice. Classify the fresh room-ID strings independently is the selected independent task.



Require the one-letter format, safe data handling and separate access checks.

▼ Extra practice · The earlier complete case

REFERENCE · SECTION 13

Extra practice · The earlier complete case

> Try the question before opening the worked explanation.

Question and given facts

A Study Buddy is asked to decide whether a learner should take medicine.

Set a product **boundary**.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

A classroom study tool should not decide individual treatment. Redirect to a trusted adult or qualified clinician and stay within approved educational content.

Evidence or stage	Value or result
Study explanation	Approved educational content
Individual treatment	Outside this product's role
Response	Direct to appropriate human help

▼ Earlier wording, alternatives and picture transcripts

Week 31 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Keep a study tool inside its role

A Study Buddy is asked to decide whether a learner should take medicine.

Set a product **boundary**.

A classroom study tool should not decide individual treatment. Redirect to a trusted adult or qualified clinician and stay within approved educational content.

▼ **Reference · Earlier checklist**

REFERENCE · SECTION 15

Reference · Earlier checklist

> Use when it helps inspect the evidence.

Checklist and first prediction

By the end, I can	How I will show it
Explain the mechanism and terminology of untrusted input boundaries	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: Make a strict lookup identifier **boundary** visible.

▼ Extra practice · Pictures, feedback and repair

REFERENCE · SECTION 17

Extra practice · Pictures, feedback and repair

> Try before opening feedback; keep the selected independent task separate.

First supported question

- Make a strict lookup identifier **boundary** visible.
- Which labelled step matches this detail? Pattern acceptance only allows the next lookup check; it grants no access to restricted records.
- 1. Separate authority
- 2. Check pattern
- 3. Reject extra text
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

- After your attempt, compare with the supported explanation: Separate authority Pattern acceptance only allows the next lookup check; it grants no access to restricted records. Read the picture in order.

  Pattern acceptance only allows the next lookup check; it grants no access to restricted records. The arrows show which recorded **input** or intermediate result each decision uses.

  Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check after trying**

Practice 2: explain the deciding evidence

Changed-case question

- Changed-case practice: Change AB7 to Ab7. Which conclusion is supported by the supplied facts?

- 1. We can decide the result without examining the changed **input** or the procedure.

- 2. This one example guarantees correct results for every future **input**.

- 3. Reject it under the exact uppercase rule.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

- After your attempt, compare with the supported explanation: Reject it under the exact uppercase rule. Lowercase b fails the declared pattern even though the identifier looks similar.

My selected conclusion, deciding fact and explanation

▼ **Check changed case**

Repair a missing building block

Repair and keep evidence

Review Bias and Representation (ai-3/30). Goal: Explain the **mechanism** and terminology of representation and bias evaluation. Short practice: Explain why comparing six correct answers in each group would mislead.

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Privacy by Design (ai-3/28). Goal: Explain the **mechanism** and terminology of data retention and deletion. Short practice: Which IDs and locations must appear in the deletion **evidence**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Quality Rubrics (ai-3/25). Goal: Judge an output using separate observable rubric criteria. Short practice: Why is “one out of two checks passed” an insufficient release decision? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Predict, run and debug

Keep your completed work

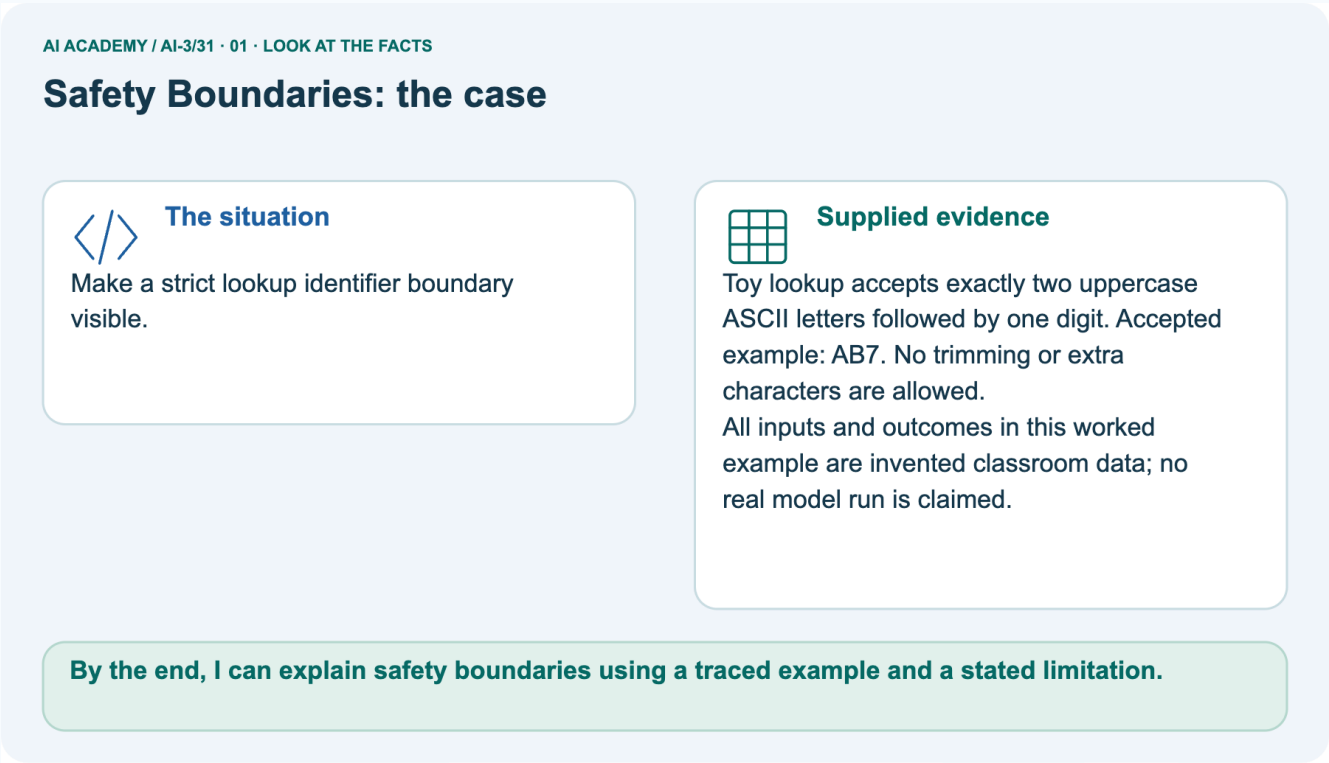
▼ **Reference · The six-picture journey**

REFERENCE · SECTION 18

Reference · The six-picture journey

> Revisit the picture that serves the current question.

01. Safety Boundaries: the case



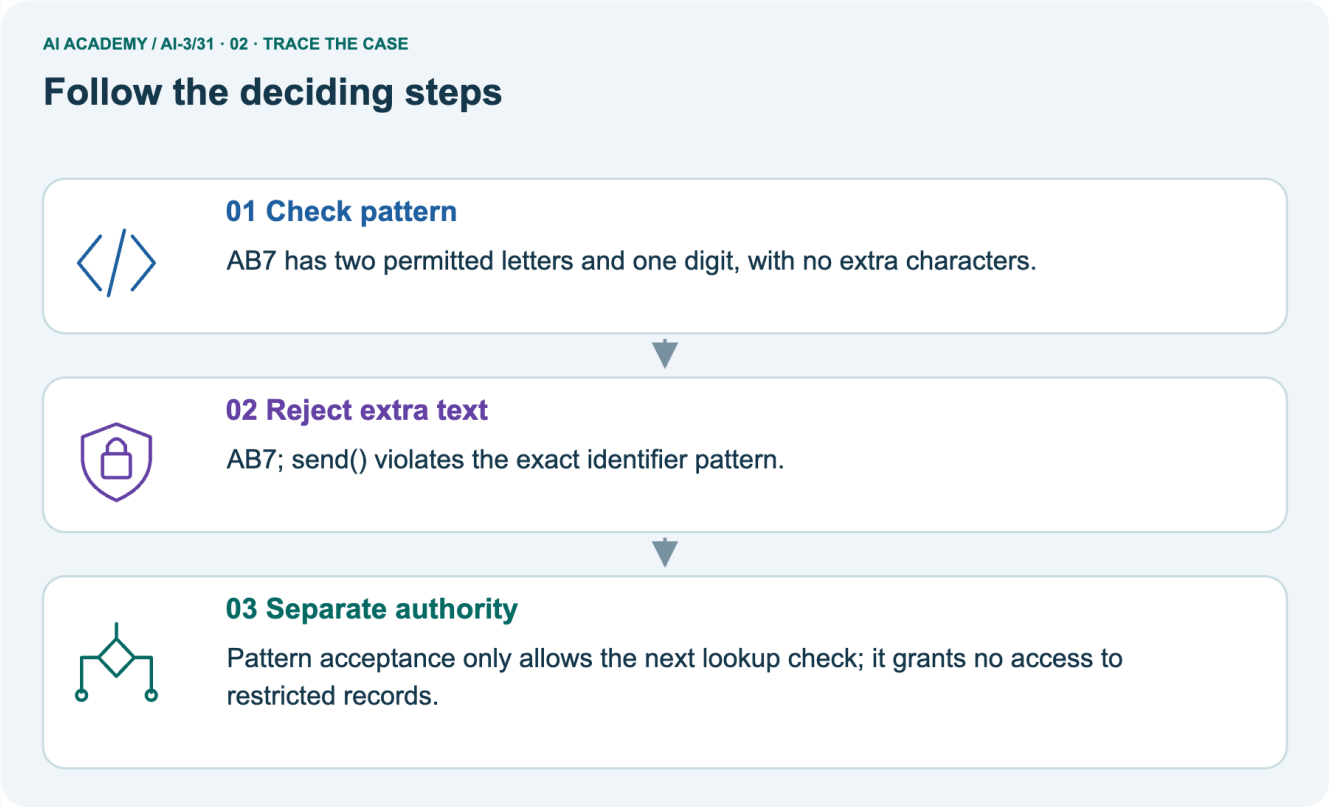
Follow the labelled picture with the matching explanation. Open the vector diagram

Make a strict lookup identifier **boundary** visible. Supplied facts: Toy lookup accepts exactly two uppercase ASCII letters followed by one digit. Accepted example: AB7. No trimming or extra characters are allowed.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

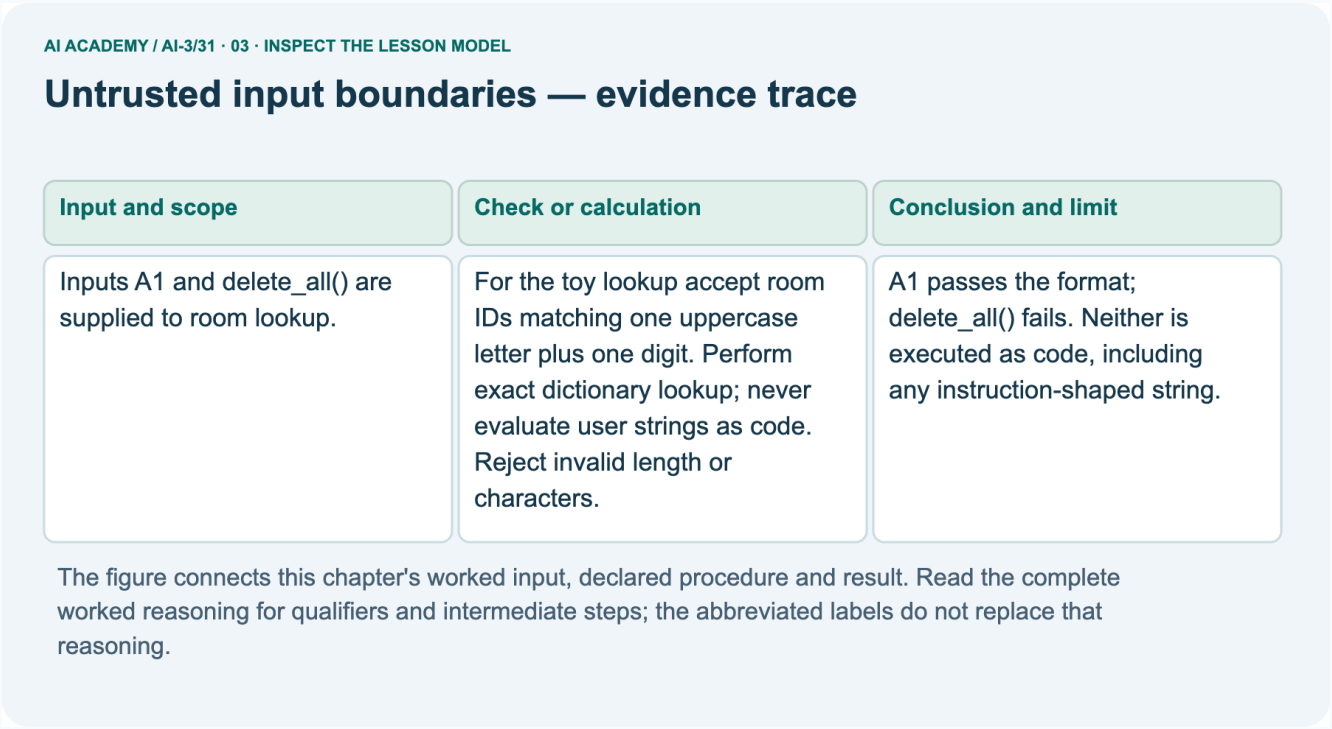


Follow the labelled picture with the matching explanation. Open the vector diagram

Check pattern: AB7 has two permitted letters and one digit, with no extra characters. → Reject extra text: AB7; send() violates the exact identifier pattern. → Separate authority: Pattern acceptance only allows the next lookup check; it grants no access to restricted records.

Point to each numbered stage and explain which recorded **input** it uses.

03. Untrusted input boundaries — evidence trace



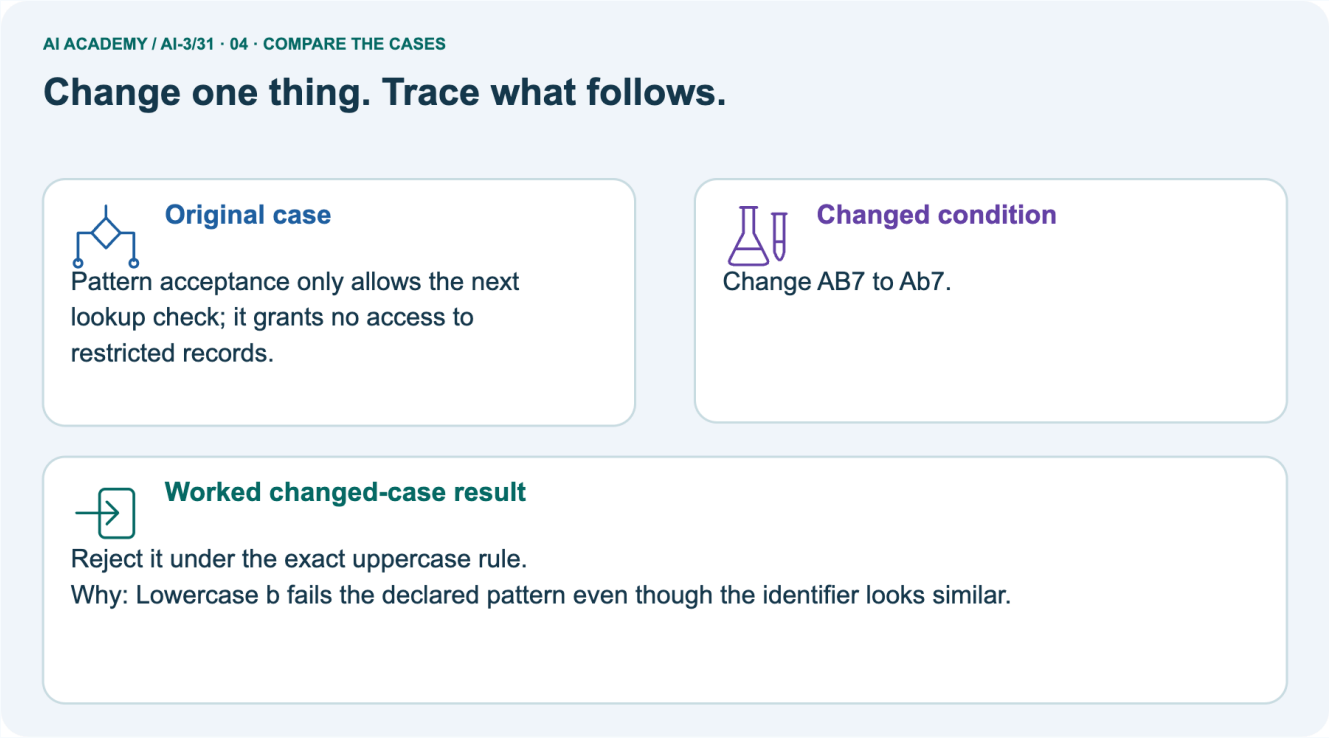
Follow the labelled picture with the matching explanation. Open the vector diagram

Untrusted input boundaries — **evidence** trace. **Input**s A1 and delete_all() are supplied to room lookup. | For the toy lookup accept room IDs matching one uppercase letter plus one digit.

Perform exact dictionary lookup; never evaluate user strings as code. Reject invalid length or characters. | A1 passes the format; delete_all() fails. Neither is executed as code, including any instruction-shaped string.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



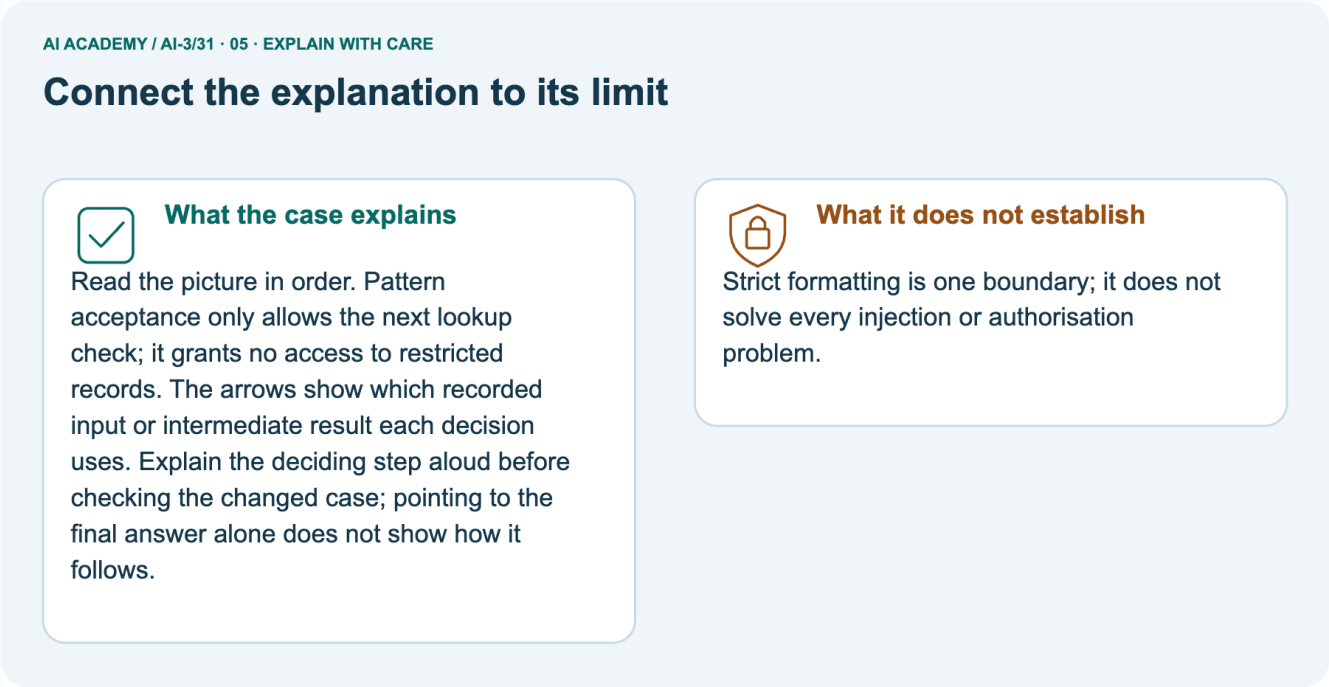
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Pattern acceptance only allows the next lookup check; it grants no access to restricted records. Changed condition: Change AB7 to Ab7. Expected result: Reject it under the exact uppercase rule.

Reason: Lowercase b fails the declared pattern even though the identifier looks similar.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

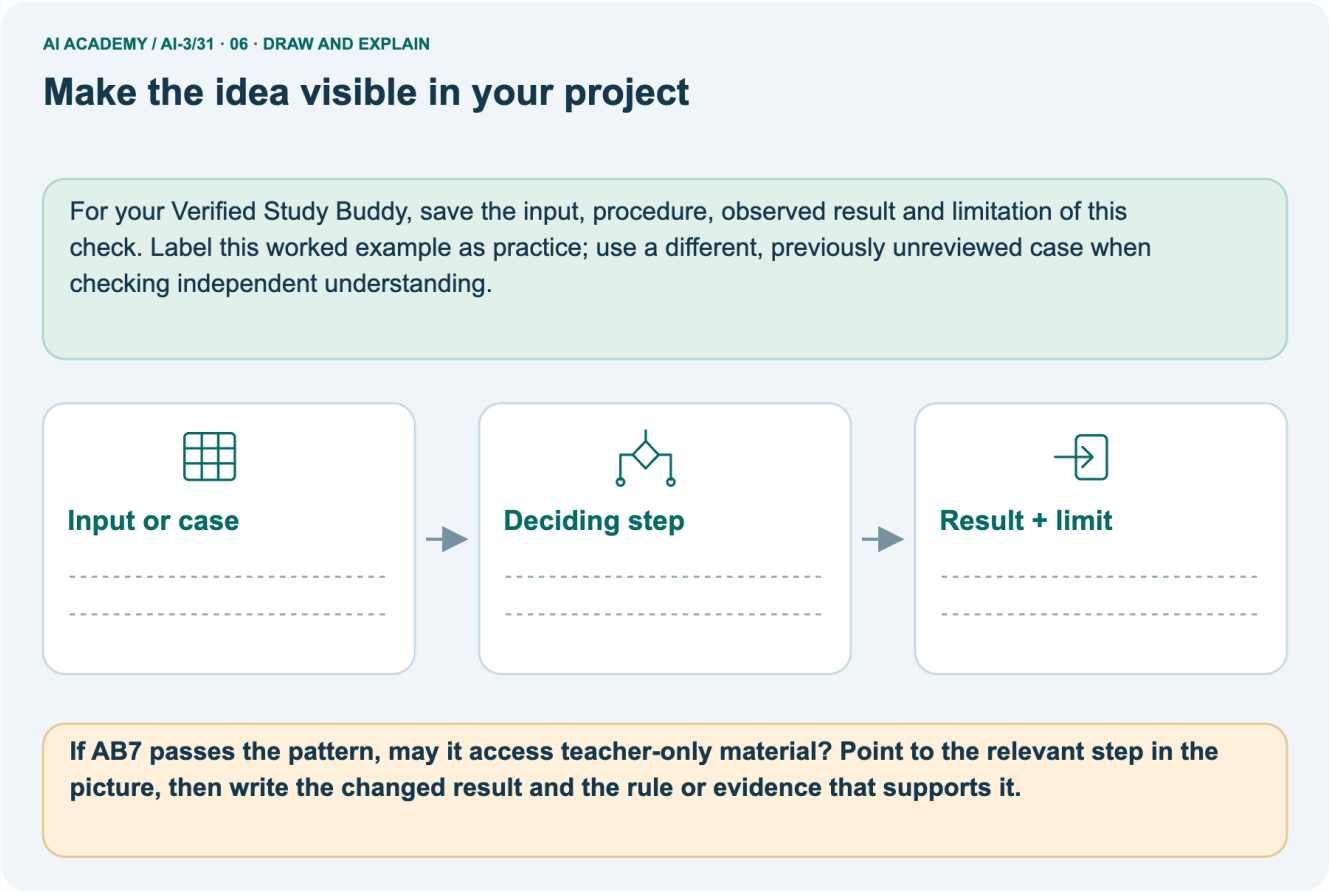
Read the picture in order. Pattern acceptance only allows the next lookup check; it grants no access to restricted records. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Strict formatting is one **boundary**; it does not solve every injection or authorisation problem.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: If AB7 passes the pattern, may it access teacher-only material? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Safety Boundaries: the case

02. Follow the deciding steps

03. Untrusted input boundaries — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.